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- [5] **Choi, T.**, O. Would, A. Salazar-Gomez, X. Liu, and G. Cielniak. Channel Randomisation: Self-Supervised Representation Learning for Reliable Visual Anomaly Detection in Specialty Crops. In: *Journal of Computers and Electronics in Agriculture*, Nov 2024. doi:10.1016/j.compag.2024.109416.
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- [8] Bandodkar, G., S. Agarwal, A. K. Sugghosh, S. Singh, and **T. Choi**. “Allot?” Is “A Lot!” Towards Developing More Generalized Speech Recognition System for Accessible Communication. In: *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI-24)*, Feb 20–27, 2024. Vancouver, Canada. doi:10.1609/aaai.v38i21.30381
- [9] **Choi, T.**, O. Would, A. Salazar-Gomez, and G. Cielniak. Self-supervised Representation Learning for Reliable Robotic Monitoring of Fruit Anomalies. In: *Proceedings of the 2022 IEEE International Conference on Robotics and Automation (ICRA 2022)*. May 23–27, 2022. Philadelphia, USA. doi:10.1109/ICRA46639.2022.9811954.
- [10] **Choi, T.**, B. Pyenson, J. Liebig, and T. P. Pavlic. Beyond Tracking: Using Deep Learning to Discover Novel Interactions in Biological Swarms. *Journal of Artificial Life and Robotics (AROB)*, Mar 2022. doi:10.1007/s10015-022-00753-y
— Extension of the *Best Paper Award* winner at the *4th International Symposium on Swarm Behavior and Bio-Inspired Robotics 2021 (SWARM 2021)*, Jun 1–4, 2021. Kyoto, Japan. Virtual event.
- [11] **Choi, T.** and G. Cielniak. Adaptive Selection of Informative Path Planning Strategies via Reinforcement Learning. In: *Proceedings of the 10th European Conference on Mobile Robots (ECMR 2021)*, Aug 31–Sep 3, 2021. Bonn, Germany. Virtual event. doi:10.1109/ECMR50962.2021.9568796
- [12] **Choi, T.**, B. Pyenson, J. Liebig, and T. P. Pavlic. Identification of Abnormal States in Videos of Ants Undergoing Social Phase Change. In: *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI-21)*, Feb 2–9, 2021. Virtual event. doi:10.1609/aaai.v35i17.17794
- [13] **Choi, T.** and T. P. Pavlic. Automatic Discovery of Motion Patterns that Improve Learning Rate in Communication-Limited Multi-Robot Systems. In: *Proceedings of the 2020 IEEE International Conference on Multisensor Fusion and Integration (MFI 2020)*, Sep 14–16, 2020. Karlsruhe, Germany. Virtual event. doi:10.1109/MFI49285.2020.9235218
- [14] Kang, S., **T. Choi** and T. P. Pavlic. How Far Should I Watch? Quantifying the Effect of Various Observational Capabilities on Long-range Situational Awareness in Multi-robot Teams. In: *Proceedings of the 1st IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2020)*, Aug 17–21, 2020. Washington, DC, USA. Virtual event. doi:10.1109/ACSOS49614.2020.00036

	[15] Choi, T., S. Kang, and T. P. Pavlic. Learning Local Behavioral Sequences to Better Infer Non-local Properties in Real Multi-robot Systems. In: <i>Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA 2020)</i>, May 31–June 4, 2020. Paris, France. Virtual event. doi:10.1109/ICRA40945.2020.9196728 [16] Choi, T., T. P. Pavlic, and A. W. Richa. Automated Synthesis of Scalable Algorithms for Inferring Non-Local Properties to Assist in Multi-Robot Teaming. In: <i>Proceedings of the 2017 IEEE International Conference on Automation Science and Engineering (CASE 2017)</i>, Aug 20–23, 2017. Xi'an, China. doi:10.1109/COASE.2017.8256320 [17] Choi, T. and H. Na. Stealthy Behavior Simulations based on Cognitive Data. <i>Journal of Korea Game Society (JKGS)</i>, 16(2):27–40, Apr 2016. doi:10.7583/JKGS.2016.16.2.27 [18] Choi, T. and H. Na. Making Levels More Challenging with a Cooperative Strategy of Ghosts in Pac-Man. <i>Journal of Korea Game Society (JKGS)</i>, 15(5):89–98, Oct 2015. doi:10.7583/JKGS.2015.15.5.89 [19] Choi, T. and H. Na. Stealthy Behavior Simulations based on Cognitive Data. In: <i>Proceedings of the 2015 IEEE International Conference on Machine Learning and Cybernetics (ICMLC 2015)</i>, 16(2):27–40, Jul 12–15 2015. Guangzhou, China. doi:10.1109/ICMLC.2015.7340900
INVITED TALKS	[20] Self-supervised Learning of Visual Anomalies in Strawberries. In: <i>International Conference on Digital Technologies for Sustainable Crop Production (DIGICROP)</i>, Mar 2022. Virtual event. [21] AI Research in Agriculture and Beyond – Successful Machine Learning under Limited Resources. In: <i>Hankyong National University, School of Computer Engineering & Applied Mathematics Seminar</i>, Dec 2021. Anseong, South Korea. [22] Local Behavior Learning for Social Temperature Prediction without Individual Ant Tracking. In: Oral Session at <i>Collective Information Processing</i>, Mar 2020. Berlin, Germany. [23] Machine Learning Applications for Video Game Development. In: <i>Hankyong National University, School of Computer Engineering & Applied Mathematics Seminar</i>, Jul 2015. Anseong, South Korea.
EXTERNAL GRANTS	Assurant - PI, “Help Desk Call Recording Analysis”, \$35,000, Jan 2026–Jun 2026. - PI, “Integrated Help Desk Recording-Ticket Analysis and Deployment Readiness”, \$40,000, Sep 2026–Feb 2027 NSF - PI, “ERI: Odor Representation Learning for Robust Detection of Co-present Foods and Pathogen Contamination Using an Electronic Nose”, \$199,995, Jul 2025–Jun 2027. Georgia Peanut Commission - PI, “Night Owl: A Low-Cost Smart Drone System for Defending Peanut Farms from Night-time Wildlife Intrusion”, \$25,000, Jul 2025–Jun 2026. USDA (through SARE) - PI, “MoCoBot: Developing a Low-Cost Night-time Mollusk Control Robot for Strawberry Growers”, Graduate Student Grants, \$21,964, Sep 2024–Aug 2026.
TEACHING EXPERIENCE	KSU, Marietta, GA, USA <i>Instructor</i> - IT 7143: Cloud Analytics - IT 7133: Enterprise AI Applications Sp 2024, Sp 2025, Sp 2026 Su 2025

- IT 7103: Practical Data Analytics **Fa 2023, Fa 2024, Fa 2025, Su 2026**
- IT 4773: Machine Learning for Enterprise **Fa 2023, Fa 2024**

ASU, Tempe, AZ, USA*Teaching Assistant*

- CSE 450/551: Design Analysis of Algorithms/Foundations of Algorithms: **Jan 2018 – May 2018**
 - Instructor: Dr. Andréa Richa
 - Graded exams, and held office hours (2 hours/week) for tutoring students.
- CSE 310: Data Structures and Algorithms: **Aug 2017 – Dec 2017**
 - Instructor: Dr. Andréa Richa
 - Taught recitation sessions (4 hours/week), graded exams, and provided instructions for C++ programming assignments.
- CSE 205: Object-Oriented Programming & Data Structures **Jan 2016 – May 2016**
 - Instructor: Dr. Xuerong Feng
 - Graded exams, and tutored students for Java programming (4 hours/week).
- CSE 100: Prin. of Programming with C ++ **Jan 2016 – May 2016**
 - Instructor: Dr. Phillip Miller
 - Supervised C++ programming laboratories (5 hours/week), and held tutoring hours (4 hours/week).
- CSE 424: Capstone Project II **Aug 2015 – Dec 2015**
 - Instructor: Dr. Debra Calliss
 - Supervised each project group with their short-term and long-term goals, and graded IT ethics essays.

MENTORING**KSU, Marietta, GA, USA**

- Cameron Redovian (Ph.D. in Computer Science) **Aug 2024 – Present**
- Zularbine Kamal (Ph.D. in Computer Science) **Aug 2026 – Present**
- Carter Corbin (M.S. in Intelligent Robotic Systems) **Jun 2026 – Present**
- Shema Nkindi Giscard (M.S. in Computer Science) **Aug 2026 – Present**
- Subeen Choi (B.S. in Industrial Systems Engineering) **Jan 2026 – Present**
- Robel Mamo (M.S. in Computer Science) **Aug 2024 – May 2026**

UCD, Davis, CA, USA

- (Team 1) Satyam Goyal, Rohan Sheth, Akash Seelam, and Kavya Sasikumar
(Team 2) Grisha Bhandodkar, Athul Krishna Sugghosh, Sahil Singh, and Shyam Agarwal (B.S. in Computer Science) **Jun 2023 – Feb 2024**
 - Serving as a mentor for the EAAI-24 Mentored Undergraduate Research Challenge
 - Advising in designing a problem, producing a solution and evaluation results, and writing a paper to submit.
- Grisha Bhandodkar, Zifei Cheng, Chonghan Wang, and Satyam Goyal (B.S. in Computer Science) **Jan 2023 – Aug 2022**
 - Directing to generate and extend DAVIS-Ag—a public image dataset of simulated plants—with labels, such as instance segmentation, bounding box, camera pose, etc., in a desirable format.
 - Advising in designing reasonable heuristic methods for active vision in agricultural environments and in training a fruit detector for validation of the DAVIS-Ag dataset.
 - Guiding while exploring embedded AI platforms like AI Habitat and AI2-THOR
- Avishai Halev (Ph.D. in Applied Mathematics) **Sep 2022 – Dec 2022**
 - Working with food scientists on efficient parameter search to build a realistic oxidation model fitting empirical datasets.

UoL, Lincoln, UK

- Owen Would (M.Sc. in Robotics & Autonomous Systems) **Mar 2021 – Sep 2021**
– Advised in studying on deep neural network-based visual anomaly detection of strawberry images particularly while designing a GAN-based method for occluding environments.

ASU, Tempe, AZ, USA

- Sehyeok Kang (M.S. in Computer Engineering) **Mar 2019 – May 2020**
– Mentored to implement Remote Teammate Localization on a physical robot platform, *Thymio*, while he was working on his thesis on correlation between prediction accuracy and observational information.
- Ricardo Weir (B.S. in Computer Science) **Mar 2018 – Dec 2018**
– Guided to develop a YOLO-based object-detection pipeline—from data annotation to model validation—to perform automated tracking of individual *Harpegnathos* ants from high-quality video recordings.

MEDIA COVERAGE

- Semafor (Sep 2025) – “It’s common scents”
- WSB Radio (Sep 2025) – “Kennesaw State researchers develop ‘Electronic Nose’ to detect food-borne illness”
- Fox 5 Atlanta (Sep 2025) – “KSU researcher developing AI-powered ‘electronic nose’”
- The Georgia Sun (Sep 2025) – “A KSU Researcher is Creating an Electronic Nose to Sniff Out Foodborne Illness”
- KSU News (Sep 2025) – “KSU researcher developing electronic nose to sniff out foodborne illness”
- KSU News (Jun 2025) – “Kennesaw State researchers developing AI-powered drones to protect peanut farms from wildlife”
- Wisconsin Ag Connection (May 2025) – “Night Vision Robot Protects Strawberry Crops”
- WSB Radio (May 2025) – “Kennesaw State University researchers develop AI robot for pest control”
- KSU News (May 2025) – “Kennesaw State researcher develops AI robot to aid farmers in fight against pests”

PROFESSIONAL SERVICE

KSU Research Champion

Aug 2025 – Present

- Co-leading the Applied Technologies Research Community, promoting collaborative research across the campus.

Workshop Organizer

- ICRA2023: “TIG-IV: Agri-Food Robotics From Farm To Fork” **Jun 2023**
– Full-day workshop on robotic innovations for agri-food systems—from farming to post-harvest processing, cooking, delivery, serving, and legislation

Conference/Journal Reviewer

- IJRR, RAM, RA-L, Biosystem. Eng., Comput. Electron Agric., IROS 2025 (Associate Editor), IROS 2023, ICRA 2023, ICRA 2022, IROS 2022, ICRA 2020

Grant Reviewer

- Research Grant at ASU GPSA **Aug 2017 – May 2018**
- Travel Grant at ASU GPSA **Aug 2016 – Jul 2017**

Conference Session Chair

- TAROS 2021 **Sep 2021**
- CASE 2017 **Aug 2017**

AWARDS	KSU	
	• Apex Award for research and scholarly excellence	Dec 2025
	KSU PechaKucha Night	
	• Best Demonstration Award	Nov 2024
	IROS2023 Workshop on Agricultural Robotics for a Sustainable Future	
	• 2nd Prize for Innovation in Precision Agriculture	Oct 2023
	SWARM 2021	
	• Best Paper Award	Jun 2021
	ASU Graduate College	
	• Completion Fellowship (\$8,550 plus tuition for 1 credit hour)	Aug 2020
WORK EXPERIENCE	ASU Ira A. Fulton Schools of Engineering	
	• Engineering Graduate Fellowship (\$700)	May 2020
	ASU School of Computing, Informatics, and Decision Systems Engineering	
	• Doctoral Fellowship (\$4,000)	Mar 2020
	ASU Social Insect Research Group	
	• Student Research Grants (\$1,550)	Nov 2018
	– Project: Deep Features for Generalizable Insect-behavior Learning.	
	SSU College of Information Technology	
	• Bronze Award at Software Development Competition	Oct 2012
	– Social Alarm: Smart Android Alarm Application (Photos & Demo)	
HARDWARE AND SOFTWARE SKILLS	Atlassian , Mountain View, CA	
	<i>Data Scientist Intern</i>	May 2018 – Aug 2018
	• Jira Duplicate Ticket Detection	
	– Designed a deep learning pipeline for human natural language to classify semantically similar tickets from customers.	
	– Gathered >124K examples to implement, train, fine-tune, and validate specialized LSTM models.	
	– Demonstrated 1) significantly higher accuracy than traditional machine learning models, 2) generalizability to the data from different sources of ticket, and 3) feasibility of similarity-based ranking scenarios.	
	Data Science & Machine Learning:	
	• PyTorch, TensorFlow, TensorBoard, OpenAI Gym, OpenCV, Scikit-learn, SciPy, NumPy, Pandas, Matplotlib, etc.	
	Programming Languages:	
	• Python, Java, C, C++, UNIX shell scripting, GNU make, MySQL, etc.	
	Operating Systems:	
	• Microsoft Windows family, Apple OS X, Linux, and other UNIX variants	
	Others:	
	• Unity 3D, MATLAB, L ^A T _E X, Git, Android, and TCP/IP programming	